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Education

2022 - ***Doctor of Philosophy*** *(Computer Science)*
School of Computing,
National University of Singapore, Singapore

2020 - 2021 ***Master of Computing*** *(Artificial Intelligence)*
School of Computing,
National University of Singapore, Singapore

2015-2019 ***Bachelor of Engineering*** *(Computer Science)*
School of Computer Science and Technology,
Xidian University, Xi'an, Shaanxi, China

Publications

2021 [1] *Self-Checking Deep Neural Networks in Deployment.*
Yan Xiao, Ivan Beschastnikh, David S. Rosenblum, Changsheng Sun, Sebastian Elbaum, Yun Lin, Jin Song Dong. (In proceedings of **ICSE 2021**) [Paper]

2020 [2] *Digraph Inception Convolutional Networks.*
Zekun Tong, Yuxuan Liang, Changsheng Sun, David Rosenblum and Andrew Lim. (In proceedings of **NeurIPS 2020**) [Paper, Poster]

Research Experiences

June. – Dec. ***Research Assistant***
2021 NUS School of Computing
*"**Evaluating the Trustworthiness of Deep Learning Systems**"*

- Advised by Prof. DONG Jin Song
- Funded by the Singapore National Research Foundation (NRF) under the National Satellite of Excellence on Trustworthy Software Systems (NSoE-TSS).
- This project will develop new techniques for determining the trustworthiness of DL systems, with an emphasis on uncovering and accounting for the underlying uncertainties.

Jan. – Jul. ***Research Intern***
2020 NUS School of Computing / NUS-Singtel Cyber Security R&D Lab
*"**Trustworthiness evaluation for deep learning systems**"*

- Advised by Prof. David S. Rosenblum

- This project is to design a white-box testing and uses the outputs of the intermediate layers in the deep neural networks to fit density distributions.
- Our work to be appear in ICSE 2021: *Self-Checking Deep Neural Networks in Deployment*.[Paper]

2019 ***Undergraduate Research Assistant,***
Data Engineering and Security Lab, Xidian Univ., Xi'an, China
"Influence Maximization Problem Meets Submodular Function Optimization"
- **Undergraduate thesis** Advised by Prof. Hui Li
- "Outstanding Undergraduate Thesis" in the School of Computer Science and Technology (Overall Score Ranked **1st/550**)

2018-2019 ***Undergraduate Research Assistant,***
Data Engineering and Security Lab, Xidian Univ., Xi'an, China
"DISCO: Influence Maximization Problem Using Deep Q-Learning and Graph Embedding"
- Advised by Prof. Hui Li
- A novel framework for IM problem that incorporates graph embedding and deep reinforcement learning techniques
- (Revised Version Submitted to **KDD 2021**) [arXiv:1906.07378]

Mar.-June. 2018 ***Research Intern & Algorithm Engineer***
JD Intelligent Cities Business Unit & JD Intelligent Cities Research, Beijing, China
"Distributed Reinforcement Learning System Research"
- Advised by Prof. Yu Zheng
- A study on distributed DQN implementation on spatiotemporal data, based on Ray execution framework.
- [Link to icity.jd.com]

Language Skills
- Mandarin Chinese
- English

Honors & Awards
2017-2018 The **First Prize** Scholarship (10%; Xidian University)
2015-2016 **Special Scholarship**, Symphony Orchestra and Chorus of Xidian University
2015-2016 The **Third Prize** Scholarship (30%; Xidian University)
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2018 **Honorable Mention** MCM/ICM-The Mathematical Contest in Modeling (Top 20% Globally)
2017 **Third Prize** "Challenge Cup" the Science and Innovation Competition, Shaanxi, China
2017 **Finalist** "Beauty of Programming" Competition, Microsoft Research Asia